Question

The following compounds can be used as monomers for chain polymerization in polymers:

ethylene, propylene, isobutylene, styrene, butadiene, methyl acrylate, vinyl chloride, vinyl acetate, methyl vinyl ether, acrylic acid, acrylonitrile

For the above substances, the following propositions are given:

1. All the above monomers can undergo free radical chain polymerization to form polymers.

2. Among the above monomers, at least four can undergo cationic polymerization to form polymers, with one of them being difficult to form high-molecular-weight polymers.

3. Among the above monomers, there are four species that can undergo anionic polymerization.

4. The number of monomers among the above that contain nitrogen atom substituents is equal to the number of monomers that can undergo all four major types of polymerization (free radical, cationic, anionic, coordination).

5. Among the above monomers, the several olefin monomers that do not contain aromatic rings can all undergo the same type of polymerization.

Analyze and determine the number of correct propositions among the above.

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

Answer

Correct Answer: **D**

Detailed Explanation

Analyze each proposition one by one:

Proposition 1: All the above monomers can undergo free radical chain polymerization to form polymers.

Counterexample: Isobutylene, whose double-bond carbons are connected to two methyl groups, exhibits significant steric hindrance and is highly prone to chain transfer reactions, making it difficult to form polymers. It primarily undergoes cationic polymerization. Therefore, this proposition is incorrect.

CHECKPOINT 1 PTS

Isobutylene has high steric hindrance and mainly undergoes cationic polymerization

CHECKPOINT 0.5 PTS

Proposition 1 is incorrect, as exemplified by the counterexample of isobutylene

Proposition 2: Among the above monomers, at least four can undergo cationic polymerization, with one of them being difficult to form high-molecular-weight polymers.

Monomers capable of cationic polymerization require substituents that can stabilize carbocations, such as isobutylene (forming stable tertiary carbocations), styrene (forming conjugation-stabilized benzyl carbocations), butadiene (forming conjugation-stabilized allyl carbocations), and methyl vinyl ether (where the oxygen atom's p-π conjugation stabilizes the carbocation).

CHECKPOINT 1 PTS

Monomers capable of cationic polymerization require substituents that can stabilize carbocations

CHECKPOINT 1 PTS

Isobutylene, styrene, butadiene, and methyl vinyl ether all have substituents that stabilize carbocations and can undergo cationic polymerization

Although propylene has an electron-donating methyl group, the resulting secondary carbocation is insufficiently stable and highly prone to chain transfer and rearrangement, leading to chain termination. Thus, it is difficult to form high-molecular-weight polymers.

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The secondary carbocation formed by propylene is not stable enough to form high-molecular-weight polymers, making Proposition 2 correct.

Proposition 3: Among the above monomers, there are four species that can undergo anionic polymerization.

Monomers capable of anionic polymerization require strong electron-withdrawing groups or conjugated systems. Examples include styrene, butadiene, methyl acrylate, and acrylonitrile.

CHECKPOINT 1 PTS

Monomers capable of anionic polymerization require strong electron-withdrawing groups or conjugated systems

CHECKPOINT 1 PTS

Styrene, butadiene, methyl acrylate, and acrylonitrile can undergo anionic polymerization

The count is four, so Proposition 3 is correct.

CHECKPOINT 0.5 PTS

There are four species capable of anionic polymerization, so Proposition 3 is correct.

Proposition 4: Among the above monomers, the number of monomers with nitrogen-containing substituents equals the number of monomers capable of undergoing all four major types of polymerization.

The only nitrogen-containing monomer is acrylonitrile. The count is one.

CHECKPOINT 1 PTS

The only nitrogen-containing monomer is acrylonitrile

Styrene and butadiene, due to their conjugated systems, can stabilize radicals, carbocations, and carbanions, and can also participate in coordination polymerization. The count is two.

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Styrene and butadiene can undergo all four types of polymerization, making Proposition 4 incorrect

Proposition 5: Among the above monomers, the non-aromatic olefin monomers can all undergo the same type of polymerization reaction.

The non-aromatic olefin monomers are ethylene, propylene, isobutylene, and butadiene.

CHECKPOINT 0.5 PTS

Non-aromatic olefin monomers include ethylene, propylene, isobutylene, and butadiene

Isobutylene cannot undergo free radical polymerization, ethylene cannot undergo cationic polymerization, and ethylene, propylene, and isobutylene cannot undergo anionic polymerization. However, for coordination polymerization, suitable catalytic systems exist for ethylene (Ziegler-Natta catalyst), propylene (Ziegler-Natta catalyst), isobutylene, and butadiene.

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The common polymerization mechanism for ethylene, propylene, isobutylene, and butadiene is coordination polymerization, making Proposition 5 correct

After analysis, the correct propositions are Proposition 2, Proposition 3, and Proposition 5. Therefore, there are three correct propositions in total.

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There are three correct propositions in total